Experiment 2

Designing and Implementing ETL Processes using Microsoft Power BI

AIM:

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

OBJECTIVES:

- To extract data from real-world business datasets using Microsoft Power BI.
- To transform and clean the extracted data using Power Query Editor.
- To apply data transformation techniques such as filtering, merging, removing duplicates, handling missing values, and changing data types
- To load the transformed data into a unified analytical model.

TOOLS / SOFTWARE REQUIRED:

• Microsoft Power BI Desktop
• Sample datasets: Global Superstore 2018.xlsx , Employee_DB.xlsx and a sample SQL database

DATASET USED: Global Superstore 2018.xlsx,Employee_DB.xlsx, Transactions.sql

THEORY:

1. **ETL** (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.

2. **Extract Phase:** The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.

3. **Transform Phase:** The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.

4. **Load Phase:** The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.

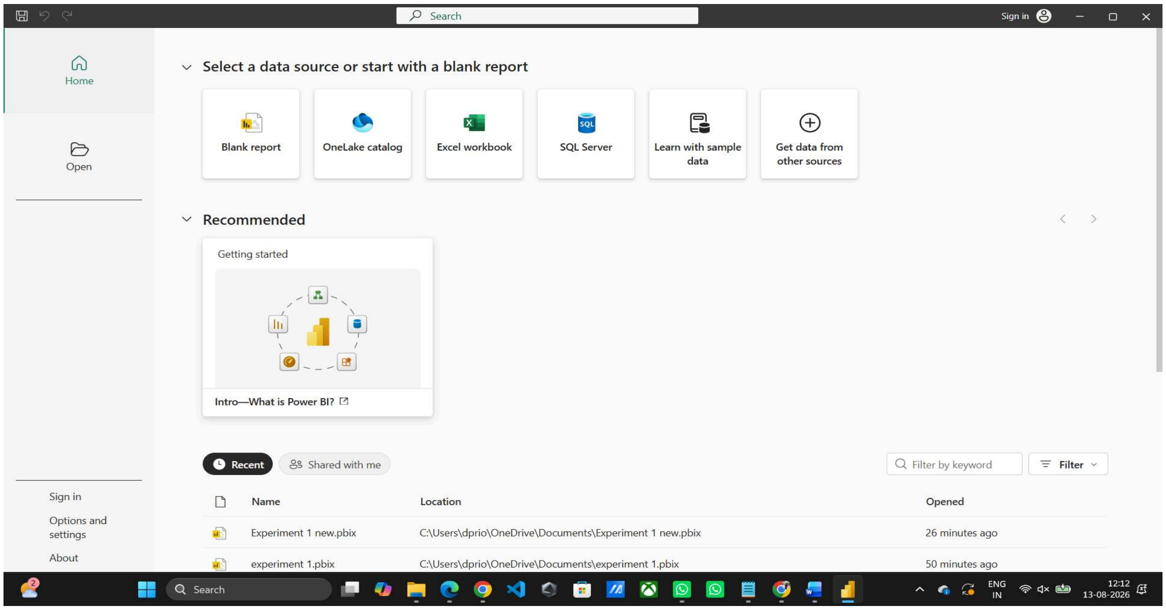
5. **Power Query Editor** serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

PROCEDURE:

Part A: Extract-Connect to Dataset

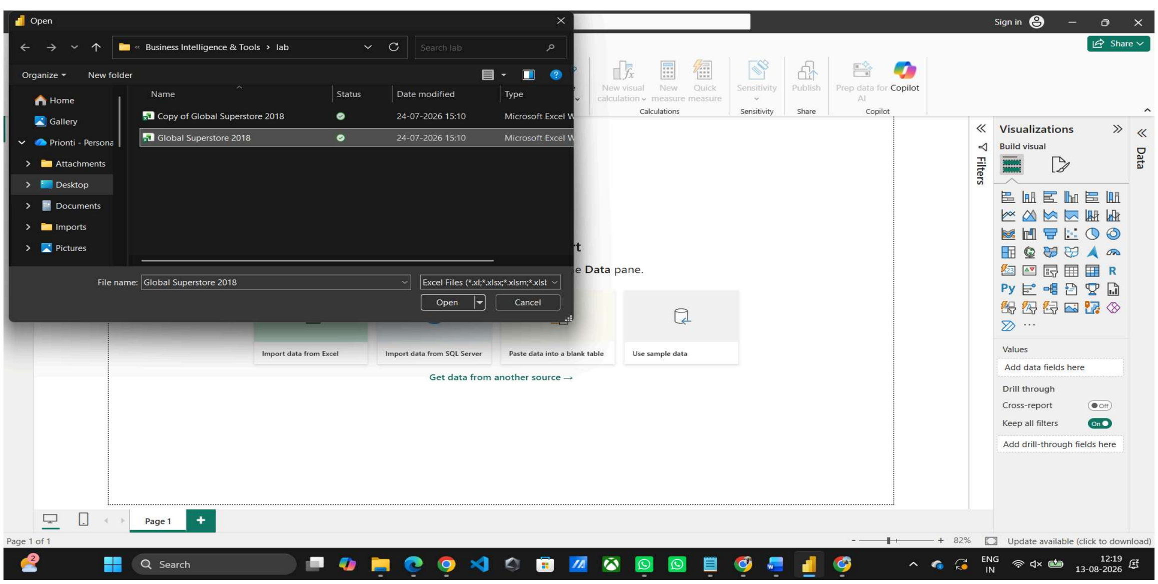
Step 1: Download the Global Superstore Dataset from Kaggle.

Step 2: Open Power BI Desktop.



Home > Get Data > Excel Workbook

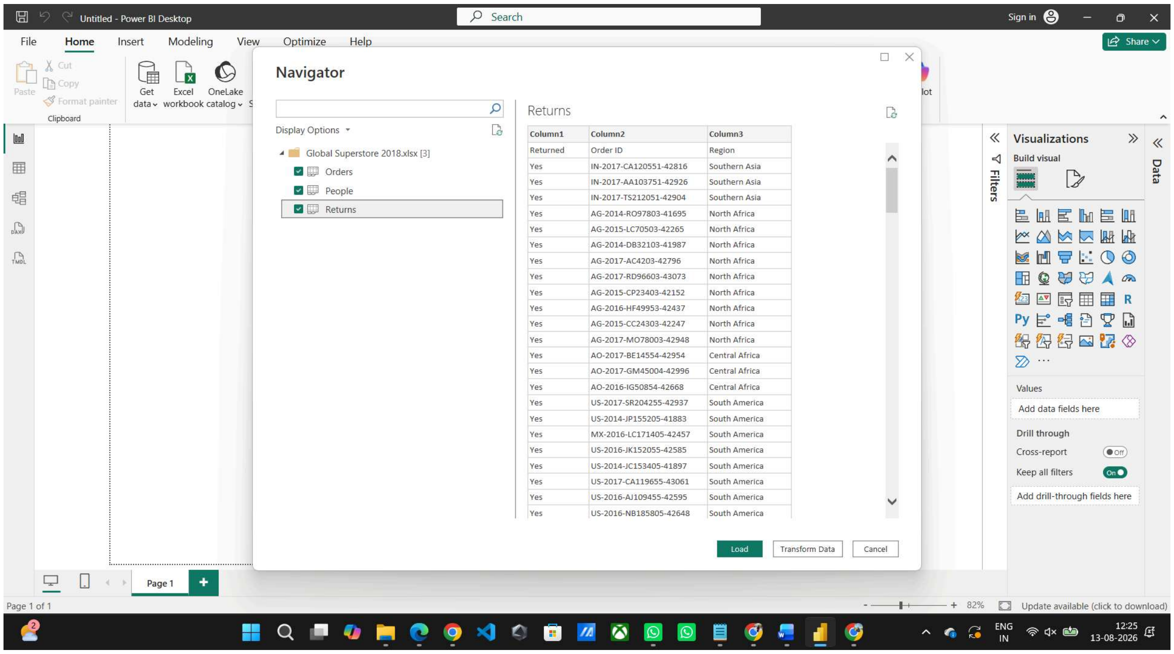
Select Global Superstore Dataset.xlsx



Click Transform Data.

Step 3: Import the following worksheets:

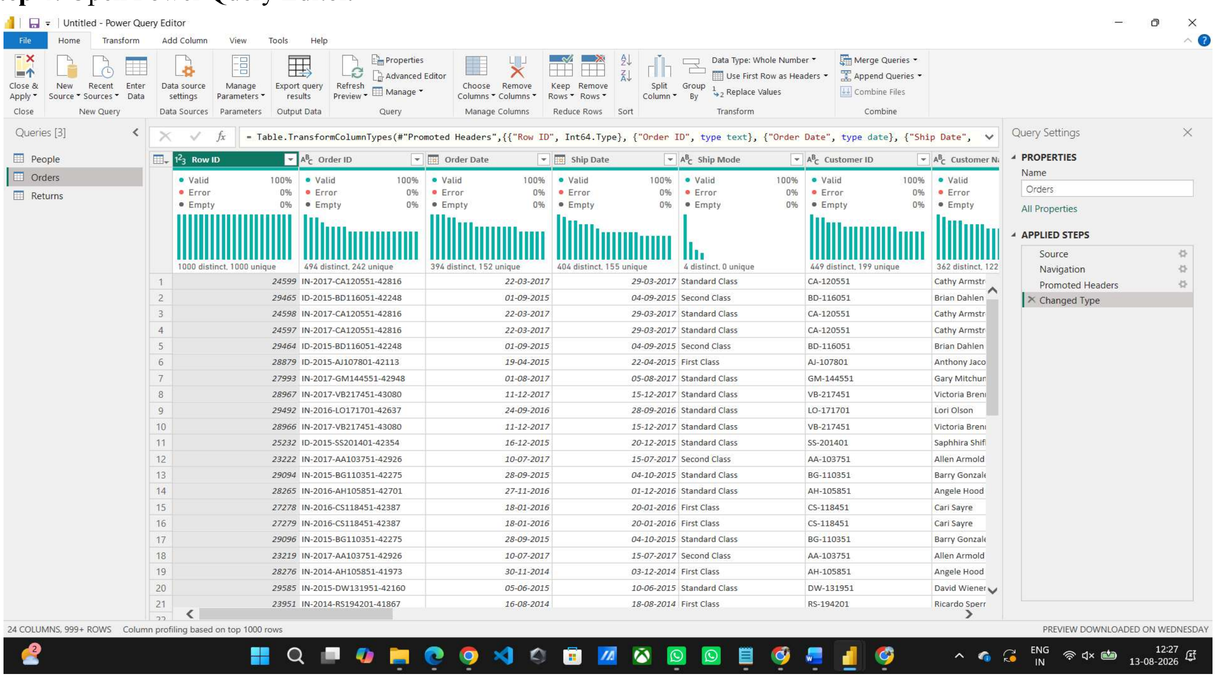
- Orders
- Returns
- People



Verify that all imported tables appear in the Queries pane.

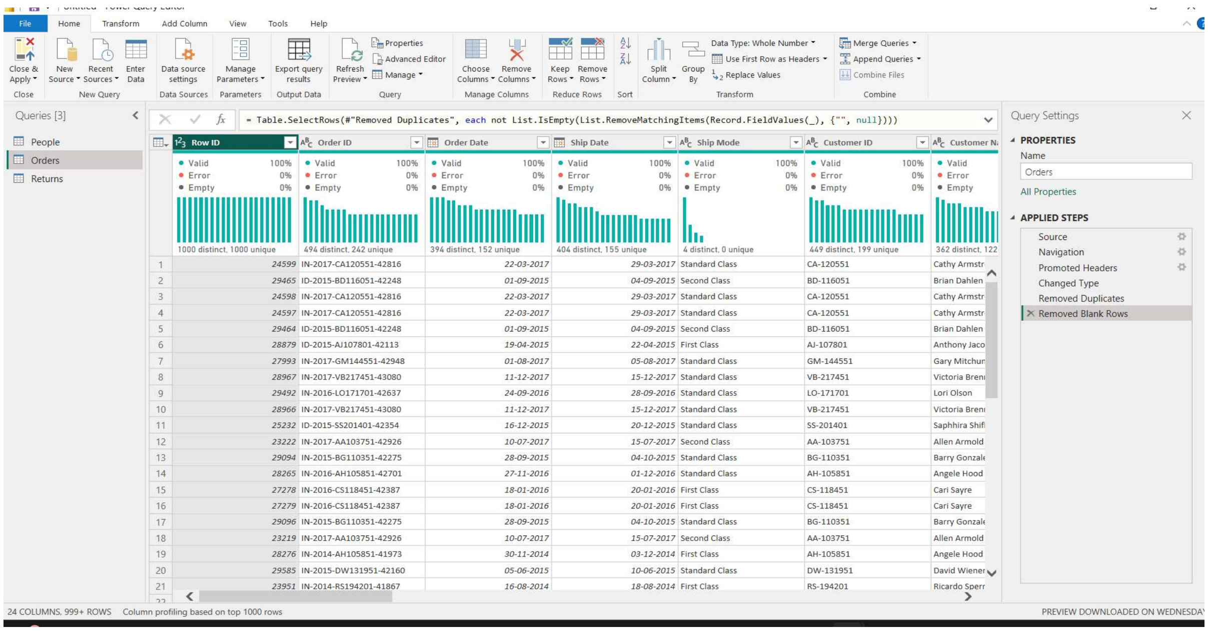
Part B: Transform-Power Query Editor

Step 4: Open Power Query Editor.

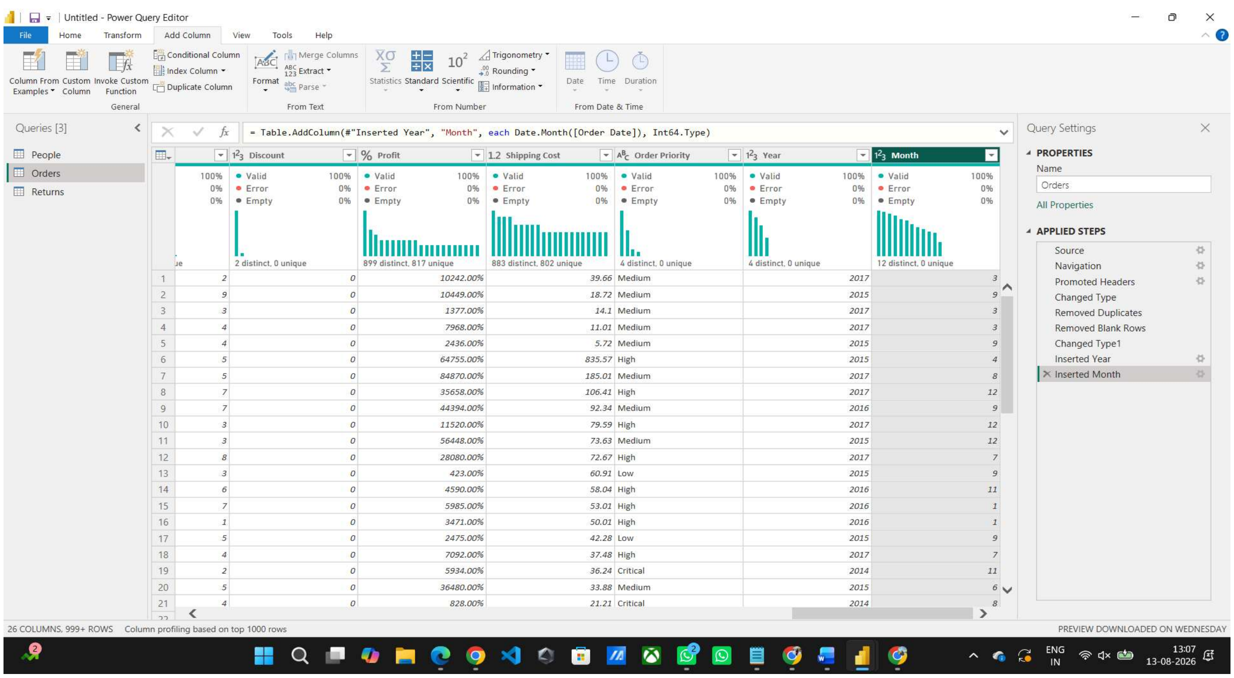


Step 5: Apply the following transformations to the Orders table:

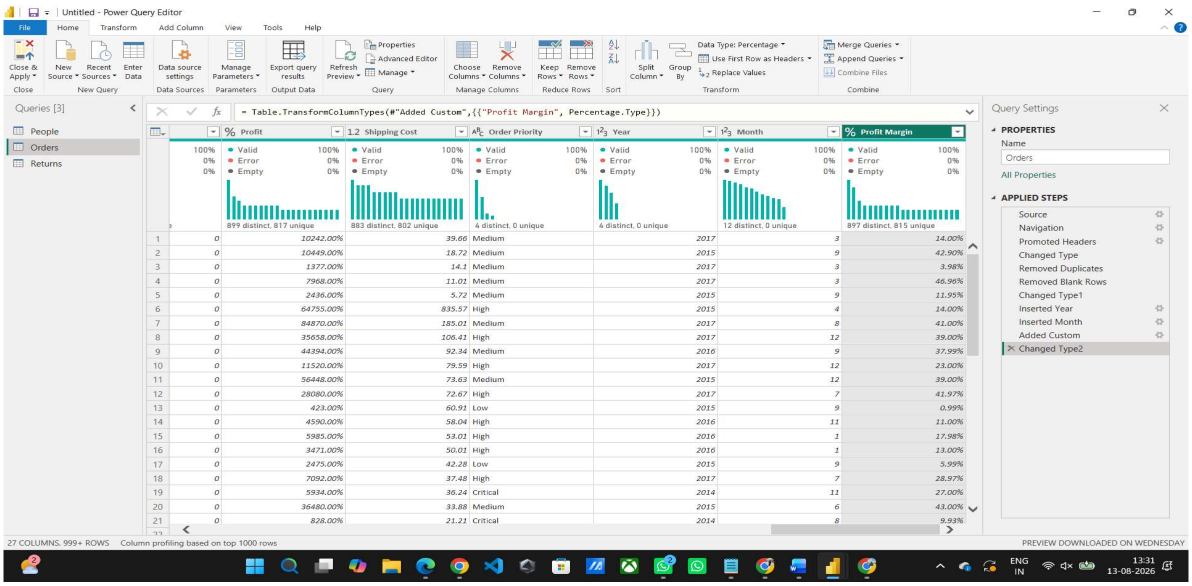
- Remove duplicate records
- Remove blank rows



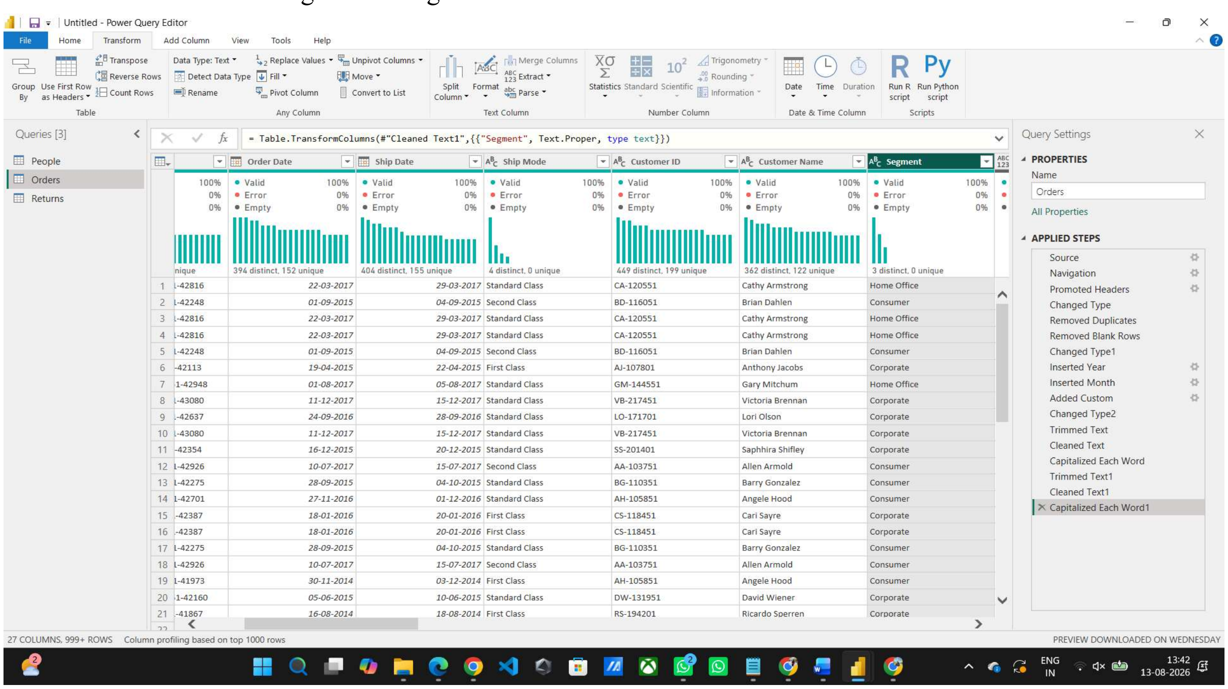
• Verify and correct data types
• Rename columns where required
• Extract Year and Month from Order Date



• Create a custom Profit Margin column

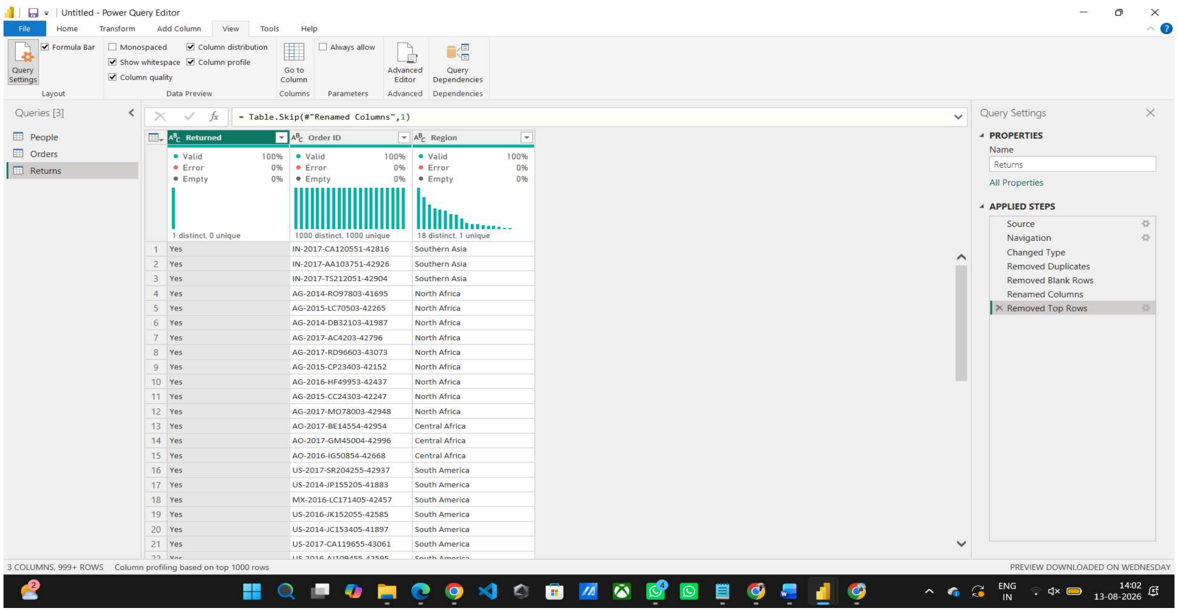


- Standardize Region and Segment values



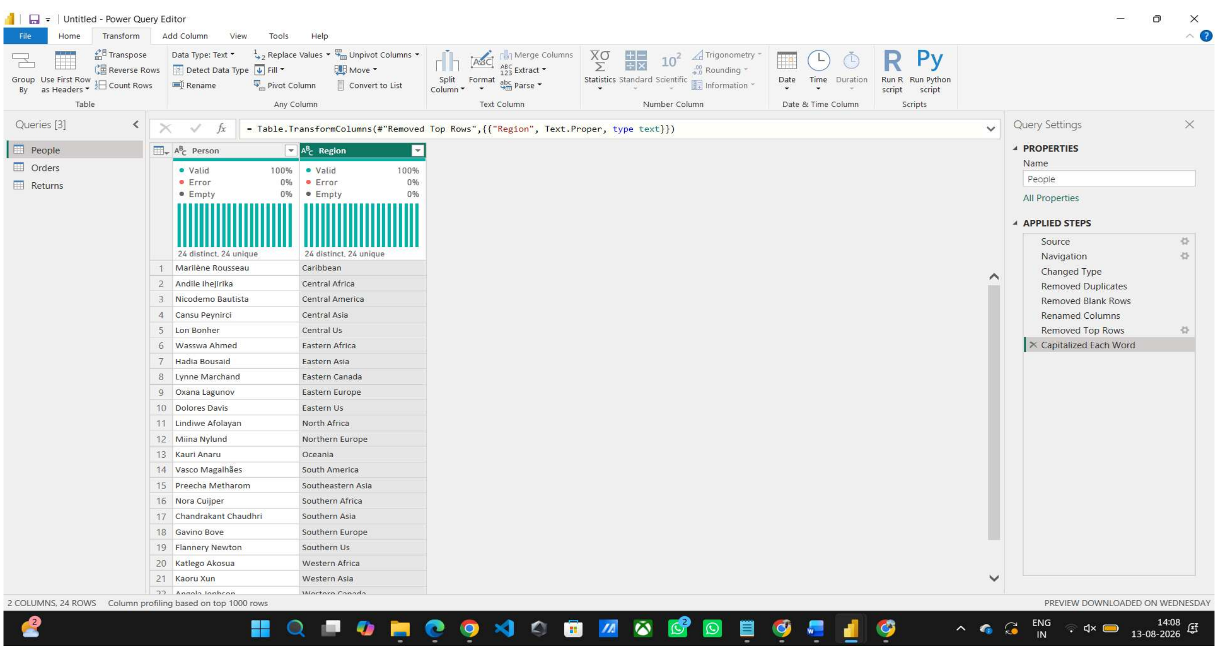
Step 6: Transform the Returns table:

- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values

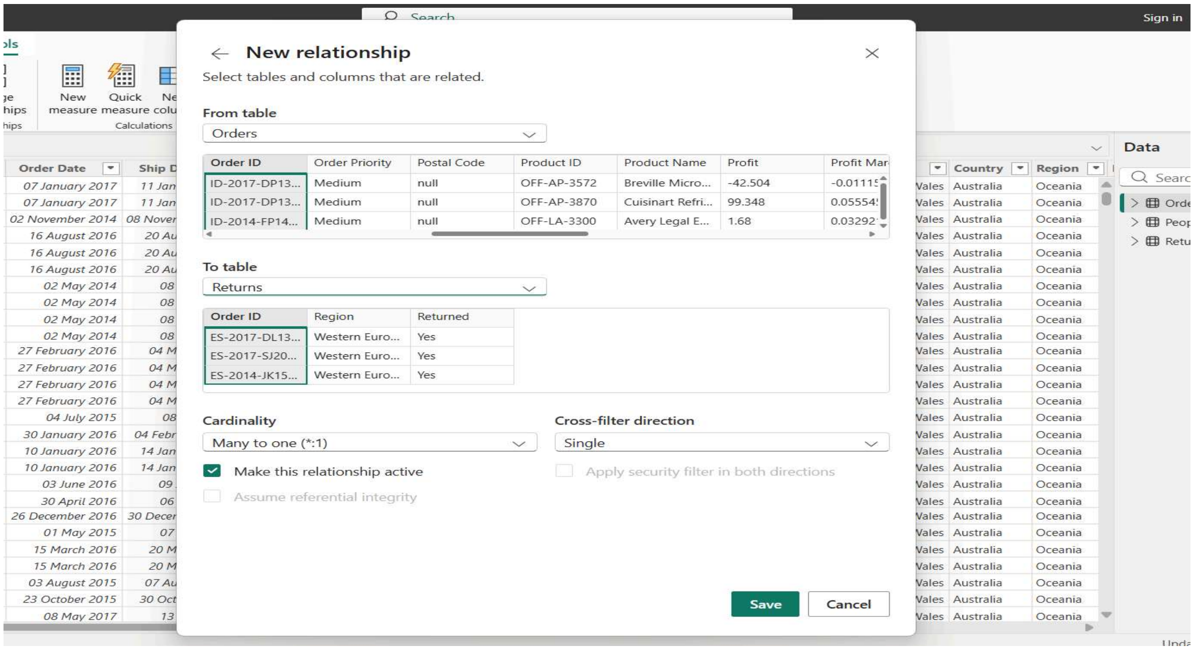


Step 7: Transform the People table:

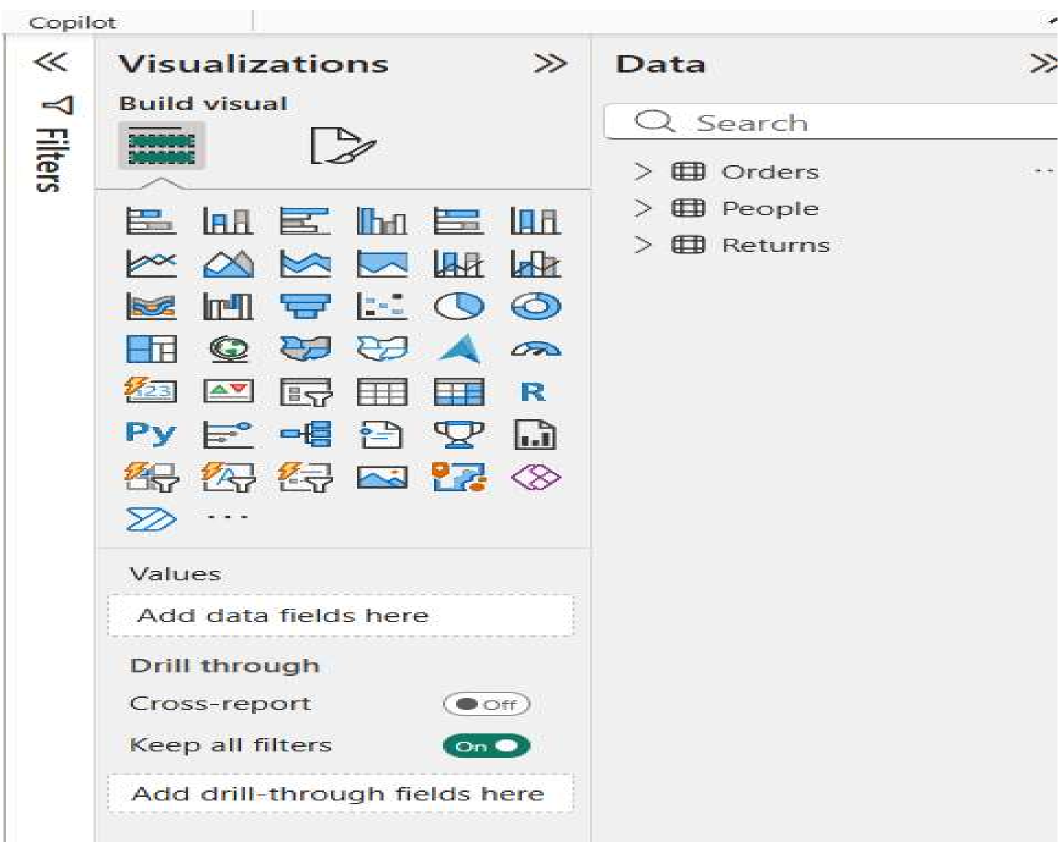
- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names



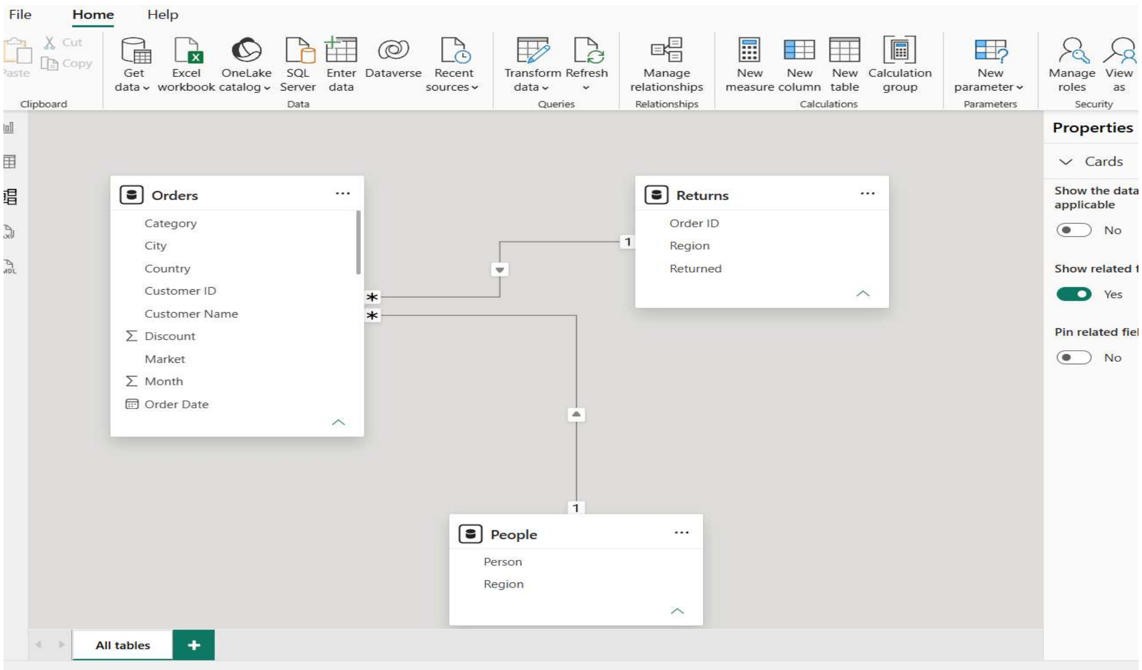
Step 8: Merge Orders and Returns tables using Order ID.



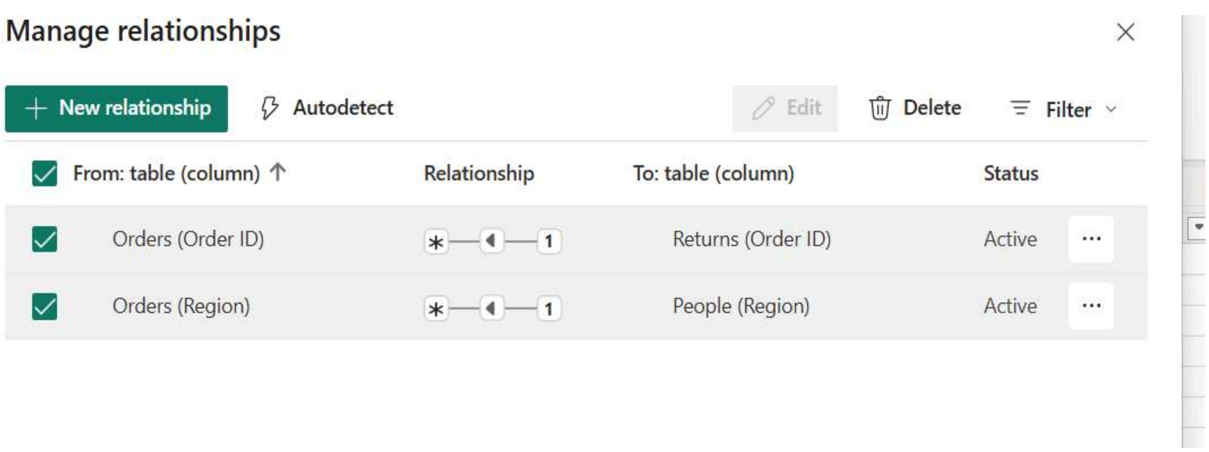
Step 9: Review all Applied Steps and click Close & Apply.



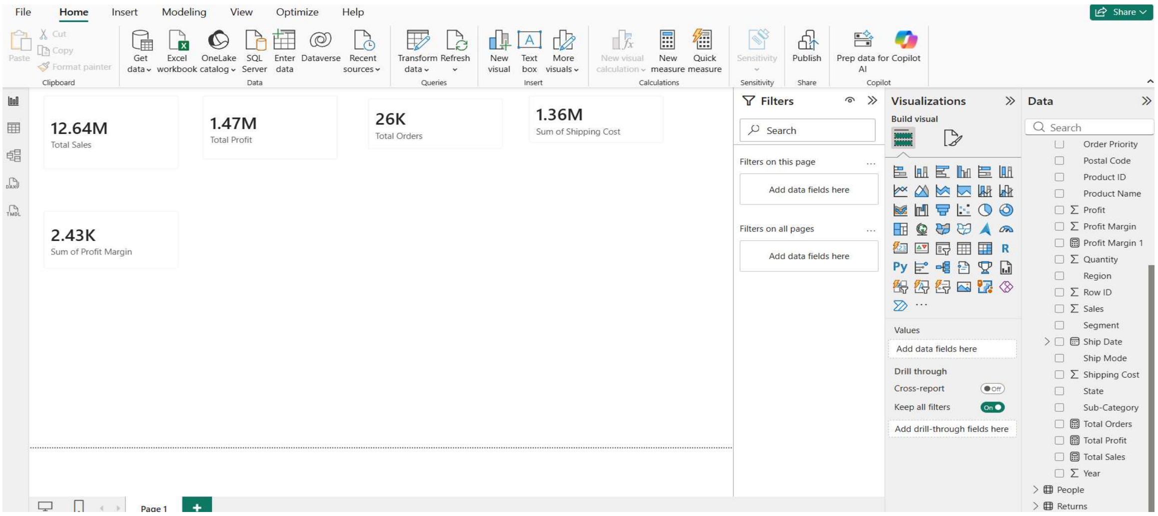
Part C: Load-Build Data Model
Step 10: Navigate to Model View.



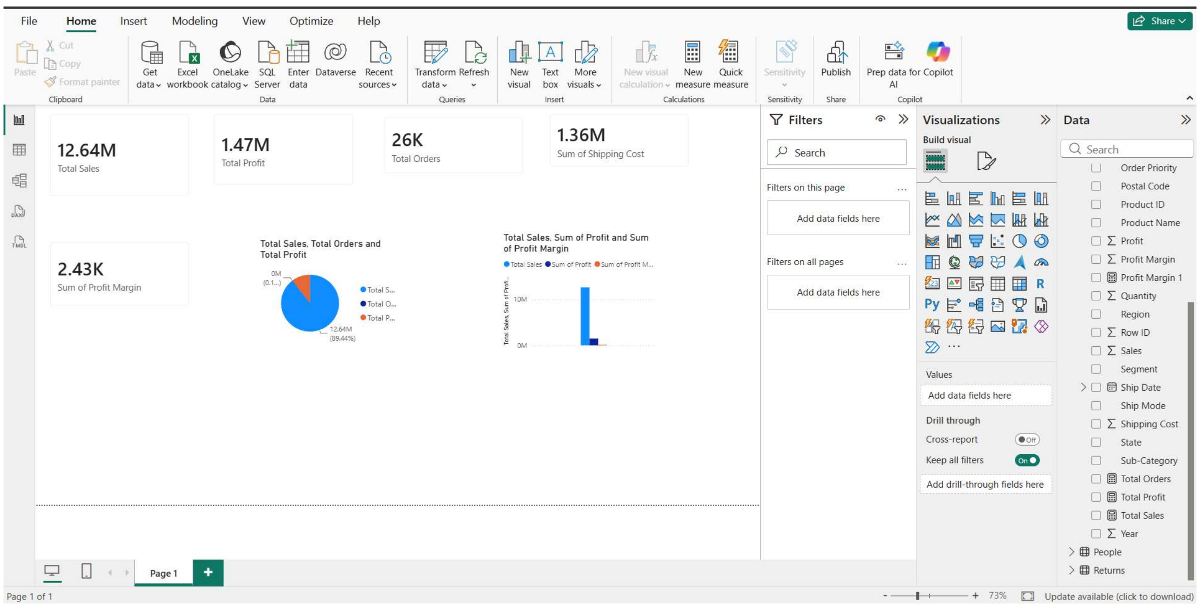
Step 11: Create relationships:
Orders ↔ Returns (Order ID)
Orders ↔ People (Region)



Step 12: Create DAX measures:
Total Sales = SUM(Orders[Sales])
Total Profit = SUM(Orders[Profit])
Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))
Total Orders = DISTINCTCOUNT(Orders[Order ID])



Step 13: Create a simple report visual using fields from related tables.



RESULT:

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

LEARNING OUTCOMES:

- Explain the ETL process and its role in Business Intelligence systems.
- Extract data from real-world business datasets using Power BI.
- Apply transformation techniques including cleaning, filtering, and merging.
- Create relationships and analytical measures in Power BI
- Validate transformed datasets for reporting and decision-making.